

Pandrosion and Free Probability: The Cauchy Transform as Per-Quotient Decomposition

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Abstract

We establish a dictionary between the Pandrosion quotient framework and Voiculescu's free probability theory. The Cauchy (Stieltjes) transform $G_\mu(z) = \int (z-t)^{-1} d\mu(t)$ of the empirical root measure $\mu_P = \frac{1}{d} \sum \delta_{\alpha_k}$ is precisely $P'(z)/(dP(z))$, which decomposes per Pandrosion quotient as $G_\mu = \frac{1}{d} \sum Q(\alpha_k, z)/P(z)$. The Pandrosion energy controls the derivative of the Cauchy transform: $E_P(z)/P(z)^2 = -dG'_\mu(z)$, identifying the tidal field (Paper 26) as d times the negative Cauchy derivative. Free cumulants are encoded in the R -transform of P'/P , and the Pandrosion root-stripping operation $P \rightarrow Q(\alpha_k, \cdot)$ is the analogue of free deconvolution.

1 The Cauchy transform of the root measure

Definition 1.1. For $P(z) = \prod_{k=1}^d (z - \alpha_k)$, the **empirical root measure** is $\mu_P := \frac{1}{d} \sum_{k=1}^d \delta_{\alpha_k}$. Its Cauchy (Stieltjes) transform is:

$$G_{\mu_P}(z) = \int \frac{d\mu_P(t)}{z-t} = \frac{1}{d} \sum_{k=1}^d \frac{1}{z - \alpha_k}. \quad (1)$$

Theorem 1.2 (Pandrosion–Cauchy identity).

$$G_{\mu_P}(z) = \frac{P'(z)}{dP(z)} = \frac{1}{d} \sum_{k=1}^d \frac{Q(\alpha_k, z)}{P(z)}. \quad (2)$$

Proof. $P'(z)/P(z) = (\log P)' = \sum 1/(z - \alpha_k)$. The per-quotient form follows from $Q(\alpha_k, z)/P(z) = 1/(z - \alpha_k)$. $\square$

The Cauchy transform of the root measure is the *per-quotient decomposition* of the logarithmic derivative.

2 The energy as Cauchy derivative

Theorem 2.1 (Energy–Cauchy identity).

$$\frac{E_P(z)}{P(z)^2} = \sum_{k=1}^d \frac{1}{(z - \alpha_k)^2} = -dG'_{\mu_P}(z) = T(z), \quad (3)$$

where $T(z)$ is the tidal field (Paper 26).

Proof. $(P'/P)' = (P''P - (P')^2)/P^2 = -E_P/P^2$. Therefore $G'_\mu = (1/d)(P'/P)' = -E_P/(dP^2)$, giving $E_P/P^2 = -dG'_\mu$. The tidal field equality $T = \sum 1/(z - \alpha_k)^2 = -\sum (1/(z - \alpha_k))' = -dG'_\mu$ follows. $\square$

Free probability object	Pandrosion form	Paper
Cauchy transform $G_\mu(z)$	$P'/(dP) = (1/d) \sum Q_k/P$	14
$-G'_\mu(z)$	$E_P/(dP^2) = T/d$	20, 26
Moments m_n	$(1/d) p_n$ (Newton sums)	21
Variance	$(1/d) \cdot E_P(m_1)/P(m_1)^2$	20

3 Free cumulants and the R -transform

The R -transform of the root measure is defined by:

$$R(w) = G_{\mu_P}^{-1}(w) - \frac{1}{w} = \sum_{n=1}^{\infty} \kappa_n w^{n-1}, \quad (4)$$

where κ_n are the **free cumulants** (related to moments by the non-crossing partition formula).

Since $G_\mu = P'/(dP)$, the R -transform is determined by the algebraic structure of $P'/P = (1/d) \sum 1/(z - \alpha_k)$. The first cumulants are:

$$\kappa_1 = m_1 = \frac{1}{d} \sum \alpha_k, \quad (5)$$

$$\kappa_2 = m_2 - m_1^2 = \text{Var}(\mu_P), \quad (6)$$

$$\kappa_3 = m_3 - 3m_2m_1 + 2m_1^3. \quad (7)$$

Remark 3.1 (Semicircular distributions). A root measure is *semicircular* (Wigner) iff $\kappa_n = 0$ for all $n \geq 3$. In this case, the R -transform is linear: $R(w) = \kappa_1 + \kappa_2 w$, and G_μ satisfies a quadratic equation. Chebyshev roots approximate the semicircular law.

4 Free convolution and root stripping

Free additive convolution $\mu \boxplus \nu$ satisfies $R_{\mu \boxplus \nu} = R_\mu + R_\nu$. In finite-dimensional approximation (Marcus–Spielman–Srivastava, 2015), if P_1, P_2 are “free” characteristic polynomials, their free convolution $P_1 \boxplus P_2$ has the root measure $\mu_{P_1} \boxplus \mu_{P_2}$.

The Pandrosion root-stripping operation $P \rightarrow Q(\alpha_k, \cdot)$ removes one atom from μ_P :

$$\mu_{Q(\alpha_k, \cdot)} = \frac{d}{d-1} \left(\mu_P - \frac{1}{d} \delta_{\alpha_k} \right).$$

This is the *free deconvolution* by a single atom, and the R -transform of $Q(\alpha_k, \cdot)$ adjusts accordingly.

5 Numerical verification

Identity	Error	Trials
$G_\mu = P'/(dP)$	$< 10^{-12}$	500
$E_P/P^2 = -dG'_\mu = T$	$< 2 \times 10^{-14}$	500
Per-quotient decomposition	exact	point-by-point

6 Conclusion

The complete Pandrosion-free probability dictionary is:

Free probability	Pandrosion
Empirical measure μ_P	Root configuration
Cauchy transform G_μ	$P'/(dP) = (1/d) \sum Q_k/P$
R -transform	Inverse of $P'/(dP)$
Free cumulants κ_n	Newton sums modulo non-crossing partitions
Free convolution $\boxplus$	R -transform addition
Free deconvolution	Pandrosion stripping $P \rightarrow Q(\alpha_k, \cdot)$
Semicircular law	$\kappa_n = 0$ for $n \geq 3$ (Chebyshev)

The Pandrosion quotient $Q(a, z) = (P(z) - P(a))/(z - a)$ is the elementary building block of the Cauchy transform: $G_\mu = (1/d) \sum Q_k/P$. This bridges the algebraic world of Papers 1–33 to the probabilistic world of Voiculescu, Speicher, and Nica.

References

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